

The Fire Guide

How to make fire from friction - the complete bow drill method, from wood selection to glowing coal

WHY THE BOW DRILL WORKS

- Friction generates heat. The spindle rotating against the hearth board creates fine wood dust that rises in temperature until it reaches ~300°C and ignites.
- The notch in the hearth board collects this hot dust (char) and concentrates it until it forms a self-sustaining coal - even after you stop drilling.
- The method works because wood is a poor heat conductor - the friction heat stays local, building until ignition.

WOOD SELECTION

- Both spindle and hearth board must be the same species or compatible soft-to-medium hardwoods.
- Best species: willow, cedar, cottonwood, mullein (stalks), elderberry, basswood, poplar. All must be bone dry.
- Avoid: pine (resin clogs the notch), hardwoods like oak or maple (too hard, friction becomes heat sink), wet or green wood of any species.
- Test: the wood should leave a faint mark but not cut like butter when dragged by your fingernail. If it squeaks, it's too hard.

SHAPING THE SET

- Spindle: 30-40cm long, 1.5-2cm diameter. One end tapered to a blunt point (goes in the hearth board). Top end rounded (goes in the handhold).
- Hearth board: flat, 2-3cm thick. Carve a shallow depression for the spindle tip. Once it burns dark, cut a pie-slice notch pointing to the edge - the notch is where coal forms.
- The bow: a flexible green branch 50-60cm long, strung with paracord or boot lace with moderate tension.
- Handhold socket: a hard stone with a natural depression, or a piece of hardwood with a socket carved in it. Lubricate with lip balm or animal fat to reduce friction at the top.

BODY MECHANICS

- Foot placement: left foot on the hearth board beside the depression, knee on the ground. Your left wrist braces against your left shin - this is your stability.
- Bow grip: wrap the cord once around the spindle. Hold the bow palm-up, elbow out.
- Stroke: long, smooth, horizontal strokes. Maintain consistent downward pressure through the handhold. Speed matters less than consistent pressure and smooth rhythm.
- Reading the smoke: thin white smoke = heating. Thick dark smoke = coal forming. When smoke continues after you stop - you have a coal.

GETTING THE COAL TO FIRE

- Tap the hearth board gently to drop the coal onto a thin piece of bark or a leaf placed under the notch before drilling.
- Tinder bundle: a bird-nest shape of fine dry grass, cattail fluff, or dry moss. The coal goes in the center; fold the bundle around it.
- Blow: long, gentle, steady breaths from 10cm away. The bundle will darken, smoke heavily, then burst into flame. Open

your hands and let it breathe.

- When it flames, build it immediately: small dry sticks teepee'd over the burning bundle, then larger sticks.

TROUBLESHOOTING

- Smoke but no coal: notch is wrong (too small, wrong angle) or wood is too hard. Deepen the notch to 1/8 of the depression diameter.
- Coal dies before transfer: you're not moving fast enough. Have bark and tinder bundle ready before drilling.
- Spindle slips: check cord tension on the bow. The spindle should rotate without slipping when you push down.
- Practicing legally at home: you can practice the full motion on a concrete floor without lighting - practice until you consistently produce smoke, then move outdoors.

>> THE SAGE'S TIP: Practice the bow drill until you can produce an ember in under 60 seconds in good conditions. At that point, you have the skill. Everything harder comes after.

The fire that kept every village alive is a skill anyone can learn. Learn it before you need it. - The Sage

